

Clinical Risk Assessment Report: Patient 14

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.42x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.3%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -1.06 [Bottom 14%] (-2.3%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 70.0 [Top 25%] (-1.4%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.05 [Avg (71th %ile)] (-1.1%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.66 [Bottom 17%] (-0.9%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.59 [Bottom 25%] (-0.7%)**
Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.04 [Top 25%] (-0.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -0.68 [Bottom 21%] (-0.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.82 [Bottom 18%] (-0.5%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Machine Learning Feature Attribution (SHAP Decision Path)

